

Analysis First-Year Exam, January 2023, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let f be a real-valued function on $[0, 1]$ and consider the statements below. Prove that (ii) implies (i); and show by example that (i) $\Rightarrow$ (ii) can fail.
 - (i) for each positive integer n the function f is Riemann-integrable on $[1/n, 1]$ and the sequence of integrals $\int_{1/n}^1 f$ converges to a real number as $n \rightarrow \infty$;
 - (ii) the function f is Riemann-integrable on $[0, 1]$.
2. Let X be a metric space on which (f_n) and (g_n) are uniformly convergent sequences of continuous real-valued functions.
 - (i) Prove that if X is compact then $(f_n g_n)$ is uniformly convergent.
 - (ii) Show that the conclusion in (i) can fail when X is not compact.
3. Let $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be continuous. By first considering polynomials in two variables, prove that

$$\int_0^1 \left(\int_0^1 f(x, y) dx \right) dy = \int_0^1 \left(\int_0^1 f(x, y) dy \right) dx.$$

4. Prove that if (f_n) is a sequence of measurable functions such that $f_n \rightarrow f$ on some measure space, then f is measurable. Hence, or otherwise, show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable then its derivative f' is Borel measurable.
5. In each of the following cases, give (if possible and with justification) a sequence (f_n) of Lebesgue integrable functions from $\mathbb{R}$ to $\mathbb{R}$ such that, where λ denotes Lebesgue measure:
 - (i) $f_n \rightarrow 0$ pointwise but $\int f_n d\lambda \rightarrow 1$ as $n \rightarrow \infty$;
 - (ii) $f_n \rightarrow 0$ pointwise but $\int f_n d\lambda \rightarrow \infty$ as $n \rightarrow \infty$.